Chapter 2: Hallucinations

Large language models sometimes *hallucinate* — to generate statements that are fluent, confident, and entirely false. Hallucinations are one of the most consequential properties of LLMs, and also one of the most misunderstood. Hallucinations are not software bugs, nor are they fixable with current models. Rather, they are an inevitable consequence of how these models work.

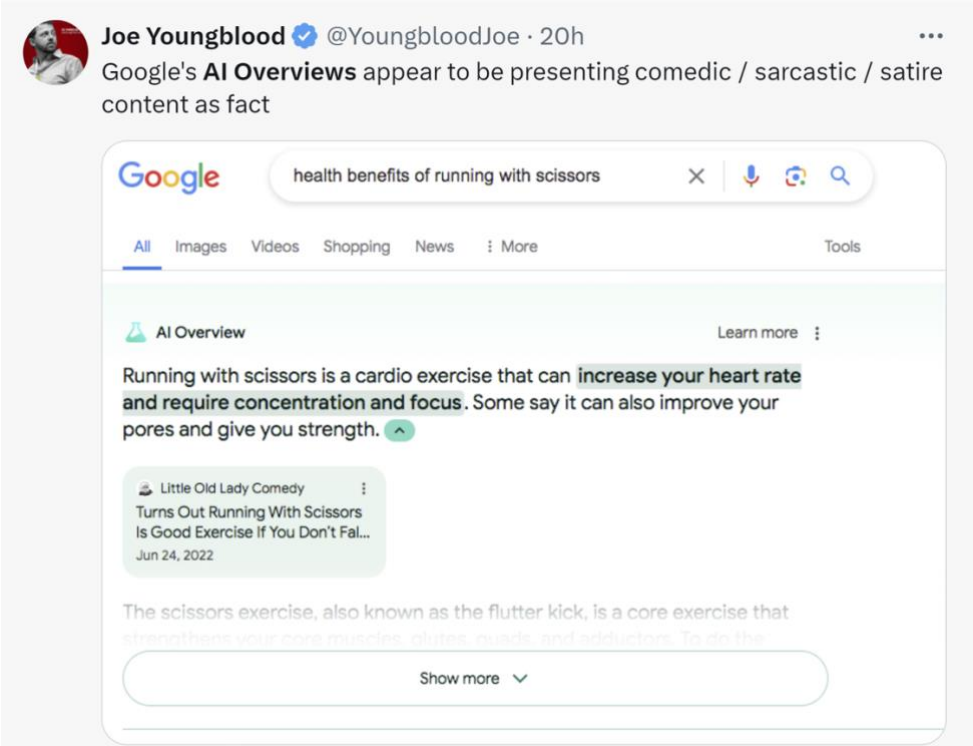


Figure 2.1. *A confidently wrong claim from Google AI Overview. From Joe Youngblood on X*

This chapter explains what hallucinations are, why they occur, and why the confident tone of fabricated output makes them especially dangerous.

2.1 Generating text that fits, not text that is true

To understand hallucinations, you must first remember what an LLM is doing when it generates a response. As discussed in the last chapter, the model is not retrieving facts from a database or reasoning from first principles or an internal model of the world.

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Rather, it is predicting a statistically plausible sequence of words given the context it has received and the data it was trained on.

This means that when you ask an LLM a factual question, it produces an answer that plausibly follows from the question — but not necessarily the correct answer.

Your experience may be that LLMs answer many questions accurately, and that's true. Given "The capital of France is," the model will overwhelmingly predict the correct next word: "Paris." That association occurs frequently in the data used to train the model, so the statistical prediction will almost certainly be right.

For this reason, questions about well-documented topics covered extensively in the training data — the plot of a famous novel, the general principles of thermodynamics, how to write a for-loop in Python — are likely to be answered correctly.

But if there is *not* a lot of information on a topic in the training data, the statistical associations may be weak and, therefore, answers will be at high risk of being hallucinations. For example, the date that an obscure law was passed, a not-so-well-known person was born, or an unremarkable event occurred may not occur frequently in training data, or several different dates may occur across training sources, leading the model to produce wrong answers to questions about it.

Statistics are another high-risk category: specific percentages, dollar figures, and counts may be not appear in the training data or appear multiple times but with different values, so the model has no mechanism to decide which reported figure is most accurate.

Citations to academic papers, court cases, news articles, or books are another high-risk category. Except for the most well-known citations, the statistical associations between the words in the citation will be weak, so the model can easily deviate from reality.

Niche technical facts are also high-risk. The more specialized the domain, the sparser the training data, and the more the model is forced to extrapolate from general patterns rather than use statistical associations derived from training data on that actual fact. A question about a well-documented programming language will likely produce accurate answers, while a question about one that's not widely used may not.

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The practical takeaway: any LLM output involving specific dates, statistics, names, citations, or technical specifications requires independent verification. The more specific the claim, and the higher the stakes are for getting it right, the more important it is to verify it.

Regardless of whether the answer is correct or fabricated, the LLMs always appears fluent and coherent, indistinguishable in form from correct output. LLMs never express doubt or give other clues that their answers are unreliable, like a person would.

It is also important to remember the stochastic nature of the process. If you ask an LLM about the date of an obscure event that has several dates associated with it in the training data, the LLM may randomly pick one of them (or come up with an entirely different date). And if you ask it a second time, it may well choose another date.

This behavior is not a limitation that can be patched away. It is a consequence of the model's fundamental architecture. Hallucinations are not a bug — a problem that will be "fixed" in the next version. They are a structural feature of how LLMs are built and trained.

Modern LLMs generally have the ability to search the web, so if you ask about an obscure fact, it may find relevant web pages and give you answers that come from there rather than strictly rely on statistical associations in the training data.

This means, however, that the truth of the statement now relies on a web page that you have not read and vetted. Figure 2.1 shows the risk of this. In that example, the LLM found a comedy website but took it as reliable and accurate information, which is most definitely is not.

2.2 The LLM does not know what it is saying

This fact has important consequences for how humans should interpret interactions with LLMs. Consider a few examples.

People who have romantic relationships with AI might ask, "Do you love me?" If the model says "yes," it is not because the model loves you. It has no feelings and is not

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sentient. It is predicting a “yes” response because that is a frequent next word in the training data in response to that question.

Similarly, if an LLM gives you an answer and you ask, “Are you sure?”, it will probably answer “yes”. This is not because it has verified the answer, but because that is the statistically likely response to that query. The model’s insistence that an answer is right should not be taken to mean that it actually is.

And if you catch an LLM in a lie and ask “Do you feel bad about lying to me?” it will reply “yes.” It does not feel bad because it does not have feelings. That is just the likely next word.

2.3 Confidence bias: fabrications delivered with certainty

One particularly dangerous aspect of LLM hallucinations is that they arrive with no warning label. When an LLM fabricates a fact, it does not lower its tone, add a hedge, or signal uncertainty. It delivers the fabrication with the same fluent confidence with which it delivers accurate information. There is no built-in uncertainty signal.

This confidence bias emerges from the human feedback used to train the LLM. Human feedback rewards clear, confident, well-organized responses, so the model learns to sound confident regardless of how much statistical support it actually has for a given claim.

For users who are not experts in the domain being discussed, there is therefore no way to detect when the model is hallucinating. A fabricated citation looks exactly like a real citation. A made-up date looks exactly like a real one. An incorrect statistic is formatted exactly like a correct one. Anytime you are asking questions that you cannot answer yourself, you have no defense against a confident hallucination.

The practical implication is that fluency and confidence are not proxies for accuracy when dealing with LLMs. A beautifully written, confidently stated response still needs verification.

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2.4 What you can do about it

This doesn't mean you should not use LLMs. It means using them in ways that keep a hallucination from turning into a consequence.

- **Verify the high-risk categories.** Specific facts (dates, statistics, names, citations, and technical specifications) should be verified. The more specific the claim, the more important it is to check it.
- **Use LLMs where the cost of a hallucination is low.** Drafting, outlining, brainstorming, summarizing text you supply, and explaining well-documented topics are tasks where an error is either easy to spot or does not matter much. A brainstormed list with one bad idea on it costs you nothing; a fabricated statistic in a report you give a client can cost you a great deal.
- **Supply the facts yourself.** Give the LLM the facts to work with — e.g., paste a source document into the conversation and ask the model to work from that text. Output grounded in material you supplied is far more reliable than output drawn from the model's memory, because the facts are now in front of it rather than reconstructed from statistical associations. The model can still misread the source, so this reduces the risk but does not eliminate it.
- **Ask for verifiable outputs, then verify them.** Get citations, URLs, case numbers, or page numbers, and then actually look them up. Fabricated references fail these tests quickly, often within seconds, and a citation that cannot be found should be treated as false.
- **Cross-check against independent sources.** Confirm important claims against a source that does not depend on the model: a textbook, an official database, a primary document, or a human expert. Asking the same model again or "Are you sure?" is not an effective check because it will regenerate an answer from the same statistical associations that produced the first one.
- **Adopt an expert-only rule for high-stakes questions.** For medical, legal, financial, or safety-related questions, don't trust an LLM unless you know enough to catch an error yourself or you will need to verify every claim before acting on it.

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If you cannot evaluate the answer and are not going to check it, you are trusting a confident voice that does not know what it is saying.

I find that the best way to use LLMs are as translators. You supply the facts, the direction, the ideas and let the LLM help you translate that to whatever format you want. That is how, for example, I wrote this book. I wrote lectures for a class and recorded the lecture delivery. I then transcribed the lectures and put them into an LLM and told it to re-write them into the format of a textbook.

In this way, I largely avoid the hallucination problem because it is just translating my lecture. However, as I edited the text, I was careful to identify facts that the LLM added and I made sure to verify those. As a result, all errors in this are mine.

2.5 Key takeaways

- LLMs generate text by predicting what word comes next based on statistical patterns, not by retrieving or reasoning about facts.
- Hallucinations are not bugs but structural consequences of how these models work.
- Errors can cascade: a wrong prediction early in a response shapes all subsequent output, compounding the inaccuracy.
- LLMs deliver fabrications with the same confident tone as accurate information — there is no built-in signal that a response is uncertain or fabricated.
- High-risk areas for hallucination include dates, statistics, citations, legal precedents, and niche technical facts. You must verify these rather than rely on the LLM.

2.6 Further reading

- Scientists invented a fake disease. AI told people it was real
- Book on Truth in the Age of A.I. Contains Quotes Made Up by A.I.

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